

COURSE TITLE : TEXTILE TESTING & QUALITY ASSURANCE
COURSE CODE : 4062
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER : 70
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Elements of Statistics, Humidity and Moisture	18
2	Fiber Testing	18
3	Yarn Testing	17
4	Fabric Testing	17
TOTAL		70

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Understand the elements of statistics & Q A D
	2	Know the influence of moisture and Humidity on textile materials
2	1	Understand the procedure of Fiber length and fineness testing
	2	Know the procedure of Fiber strength, maturity and trash testing
3	1	Understand the procedure of Yarn count and strength testing
	2	Know the procedure of Yarn twist, Yarn evenness, Yarn faults and Yarn hairiness testing
4	1	Understand the method of testing of Fabric dimension, threads density, crimp and strength
	2	Know the estimation of Fabric abrasion, stiffness, handle, air and water permeability

SPECIFIC OUTCOME

MODULE I ELEMENTS OF STATISTICS, HUMIDITY AND MOISTURE

- 1.1.0 Understand the elements of statistics & Q A D**
 1.1.1 Define Textile testing
 1.1.2 State the Objectives of Textile Testing
 1.1.3 Compare Quality control and Quality assurance

- 1.1.4 Define statistics
- 1.1.5 Explain the methods of sampling – random and biased sampling
- 1.1.6 State the meaning of population and individual in statistics
- 1.1.7 Illustrate the Construction of Frequency distribution
- 1.1.8 Draw the different types of Graphical representation of frequency distribution
- 1.1.9 Draw the different types of frequency curves
- 1.1.10 Determine the measures of central Tendency – mean, median and mode
- 1.1.11 Estimate various measures of dispersion – Range, Mean deviation, PMD, SD and CV%

- 1.2.0 Know the influence of moisture and Humidity on textile materials**
- 1.2.1 State the Importance of Humidity
- 1.2.2 Define the expressions of humidity – Absolute humidity and relative humidity
- 1.2.3 List the instrument used for measuring the humidity of the atmosphere
- 1.2.4 Explain the working of sling hygrometer
- 1.2.5 State the features of electronic hygrometer
- 1.2.6 Define standard atmospheric conditions
- 1.2.7 Define Moisture content, regain and standard regain
- 1.2.8 Explain the working of conditioning oven
- 1.2.9 Describe the features of electronic moisture analyzer
- 1.2.10 List the standard regain of important textile materials
- 1.2.11 Estimate the standard regain of blends – P/C, P/W, P/V and P/V/W
- 1.2.12 Define the term moisture hysteresis

MODULE II FIBER TESTING

- 2.1.0 Understand the procedure of Fiber length and fineness testing**
- 2.1.1 List the physical properties of Textile fibers
- 2.1.2 State the importance of fiber length
- 2.1.3 Define the length parameters
- 2.1.4 Explain the procedure of fiber length testing by Baer sorter
- 2.1.5 Describe the features of Digital Fibrograph
- 2.1.6 State the importance of fiber fineness
- 2.1.7 Describe the testing procedure of Sheffielded micronaire instrument
- 2.1.8 Explain the working of ATIRA fiber fineness tester

- 2.2.0 Know the procedure of Fiber strength, maturity and trash testing**
- 2.2.1 State the importance of fiber strength
- 2.2.2 Define the important strength terminologies
- 2.2.3 Draw the stress-strain curve & define young's modulus, yield point, yield stress and Yield strain
- 2.2.4 Draw the stress-strain curve of important textile fibers
- 2.2.5 Illustrate the elastic properties
- 2.2.6 Describe the working of Stelometer
- 2.2.7 State the importance of maturity
- 2.2.8 List the different methods of determination of maturity
- 2.2.9 Explain the testing procedure of maturity of cotton by caustic soda swelling method
- 2.2.10 Estimate the FQI and its relation with CSP

- 2.2.11 State the features of AFIS
- 2.2.12 List and explain the features of HVI
- 2.2.13 Define the term cleaning efficiency, lint, trash and invisible loss
- 2.2.14 Explain the working of Shirley trash analyzer

MODULE III YARN TESTING

3.1.0 Understand the procedure of Yarn count and strength testing

- 3.1.1 List the properties and characteristics of yarn
- 3.1.2 State the importance of yarn count
- 3.1.3 Describe the working Knowles balance
- 3.1.4 Explain the working of Beesley's balance
- 3.1.5 State the importance of yarn strength
- 3.1.6 List the factors affecting yarn strength
- 3.1.7 State the Principles of tensile strength testing
- 3.1.8 Explain the working of single yarn strength tester with CRT principle
- 3.1.9 Explain the working of single yarn strength tester with CRL principle
- 3.1.10 Describe the working of lea strength tester

3.2 .0 Know the procedure of Yarn twist, Yarn evenness, Yarn faults and Yarn hairiness testing

- 3.2.1 State the importance of yarn twist
- 3.2.2 Define twist, Z-twist, S-twist, Twist on twist and weft on twist
- 3.2.3 List the different methods of twist determination
- 3.2.4 Explain working of tension type yarn twist tester
- 3.2.5 Describe the working of Take-up twist tester
- 3.2.6 State the importance of evenness of yarn
- 3.2.7 Define the important terms used in yarn evenness
- 3.2.9 List the methods of determination of evenness—cutting & weighing, visual and electronic
- 3.2.10 Explain the Yarn appearance grading by ASTM standard board
- 3.2.11 Explain the features of Uster evenness tester
- 3.2.12 Classify the basic classimat faults
- 3.2.13 Define yarn hairiness
- 3.2.14 Explain the working of Shirley hairiness tester

MODULE IV FABRIC TESTING

4.1.0 Understand the method of testing of Fabric dimension, threads density, crimp and strength.

- 4.1.1 List the fabric properties and characteristics
- 4.1.2 State the Dimensional properties of Fabric
- 4.1.3 Explain the working of fabric thickness tester
- 4.1.4 Find the cover factor of the fabric
- 4.1.5 Define crimp, crimp percentage and crimp amplitude
- 4.1.6 Describe the working of Shirley crimp tester

- 4.1.7 Define various types of fabric strength - tensile, tearing and bursting strength
- 4.1.8 Explain the working of Fabric tensile strength tester
- 4.1.9 Differentiate between raveled strip, cut strip, and grab test method
- 4.1.10 Describe the working of Elmendorf Tearing strength tester
- 4.1.11 Explain the testing of bursting strength by Hydraulic bursting strength tester
- 4.2.0 Know the estimation of Fabric abrasion, stiffness, handle, air and water permeability**
 - 4.2.1 List the Types of abrasions
 - 4.2.2 Determination of abrasion by Martindale abrasion tester
 - 4.2.3 Define pill and state the causes of pill formation
 - 4.2.4 Determination of fabric stiffness by Shirley stiffness tester
 - 4.2.5 Define handle properties of the fabric
 - 4.2.6 Define Drape and drape co-efficient
 - 4.2.7 Determination of crease recovery by Shirley crease recovery tester
 - 4.2.8 Define air and water permeability
 - 4.2.9 Determination of air permeability by Shirley air permeability tester
 - 4.2.10 Determination of water permeability by Bundesmann water permeability tester

CONTENT DETAILS

MODULE - I

ELEMENTS OF STATISTICS, HUMIDITY AND MOISTURE

Definition of Textile Testing – Objectives of Textile Testing - Comparison of quality assurance and quality control - Definition of statistics - Selection of Sample for testing - population and individual in statistics - frequency table – Graphical representation of frequency distribution –Histogram, frequency polygon and frequency curve - types of frequency curves- Normal, Asymmetric, bi-modal, U-shaped - Measures of central Tendency – mean, median and mode - Measures of Dispersion – Range, Deviation, standard deviation, CV% - Humidity - expressions of humidity – Absolute humidity and relative humidity - Measurement of Humidity of the atmosphere by sling hygrometer – features of electronic hygrometers - Standard atmospheric condition, standard atmosphere and standard testing atmosphere – Expression of moisture in a textile material - Estimation of Moisture content and regain of Textile materials by conditioning oven and electronic moisture meter - Standard moisture regain of important textile fibers - Estimation of standard regain of a blend P/C, P/W, P/V and P/V/W- Moisture Hysteresis

MODULE - II

FIBER TESTING

Properties of Textile fibers - Importance of fiber length - Fiber length parameters- EL, ML, SF, Dispersion, Span length, 2.5% and 50% span length, uniformity ratio - determination of fiber length parameters by Baer sorter and Digital fibrograph - Importance of fiber fineness - Determination of fiber fineness by Sheffield Micronaire and ATIRA fiber fineness tester - Importance of fiber strength - Strength terminologies-Load, Breaking load, Stress, Mass stress, Tenacity, Strain, Elongation, Extension, Breaking extension, Load-elongation curve, stress-strain curve, young's modulus, yield point, yield stress, yield

strain - Stress-strain curve for cotton, wool, silk, viscose rayon, nylon, polyester - Elastic properties- elastic recovery, elastic extension and permanent set - Estimation of fiber strength by Stelometer - Importance of fiber maturity - List the different methods of determination of fiber maturity - Estimation of fiber maturity by Caustic soda swelling method - Importance of FQI and its relation to lea CSP – Features of Advanced Fiber Information System [AFIS] - Features of High Volume Instrument [HVI] – Cleaning efficiency, Lint, Trash and invisible loss - Estimation of trash content in cotton by Shirley trash analyzer

MODULE - III

YARN TESTING

Properties and characteristics of yarn - Importance of yarn count - Determination of yarn count by Knowles balance and beesley's balance - The Importance of yarn strength – List the factors affecting yarn strength - Principles of tensile strength testing-CRL. CRE, CRS and CRT - Estimation of single thread strength by CRT and CRL principle - lea strength tester - importance of Yarn twist - Definition of twist, S twist, Z twist, twist on twist, weft on twist – Different methods of twist determination - Estimation of twist in yarn by tension type twist tester and take-up twist tester- Importance of evenness of the yarn – terms in evenness – random variation, periodic variation, wave length, amplitude, short, medium and long term period variations -Methods of determination of evenness - Yarn appearance grading by ASTM standard boards – Uster evenness tester - Classification of basic classimat faults – yarn hairiness - Determination of yarn hairiness by Shirley hairiness tester

MODULE - IV

FABRIC TESTING

Fabric properties and characteristics - Dimensional properties of Fabric - Threads density - Fabric weight - Count of the yarn used for fabric manufacture - Determination of cover factor – Warp cover factor, Weft cover factor and Cloth cover factor - Crimp in yarn - Crimp percentage and crimp amplitude - Shirley yarn crimp tester - Definition of pill and causes of pill formation - Fabric strength - Determination of Fabric tensile strength by tester working on CRT principle - Elmendorf Tearing strength tester - Hydraulic bursting strength tester - Fabric abrasion resistance - Types of abrasions - Martindale abrasion tester - Fabric stiffness - Bending length, flexural rigidity and bending modulus - Cantilever principle - Shirley stiffness tester - Drape and drape co-efficient - Determination of drape by Drape meter - Crease property - Crease resistance and Crease recovery - Shirley crease recovery tester - Define air permeability - air permeability, air resistance and air porosity - Shirley air permeability tester - Water permeability and water repellency - Bundesmann water permeability tester

REFERENCE BOOKS

1. J.E. BOOTH Principles of Textile Testing Butterworth third edition
2. Grover Hand book of Quality control and Textile testing Butterworth
3. SITRA Principles of Textile Testing
4. Alagappan & Gopalakrishnan - Textile Testing SSM institute of technology
5. ISO Hand book of textile testing